A CASE-STUDY BASED REPORT

ON

Malicious Traffic Pattern Clustering Using K-Means and Hierarchical Algorithms on CICIDS2017 Network Flow Data

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Signature of the Team members

1.Abstract

Network intrusion datasets are usually analysed with supervised classifiers that need labelled attacks. This case study asks a different question: when flow-level traffic records are grouped without looking at their labels, do the resulting clusters correspond to attack types? Using a stratified sample of the CICIDS2017 flow data, seven flow-level features were selected, cleaned, standardised and clustered with K-Means and Agglomerative (Hierarchical) clustering using k = 4. Cluster quality was measured with the Silhouette score, Davies-Bouldin index and Calinski-Harabasz index; K-Means scored better on all three (0.384, 0.827 and 5514.5 against 0.348, 0.954 and 5102.3). A three-component PCA projection, which explains 60.4% of the variance, was used for visualisation. The central finding is a negative but honest one: the clusters mainly reflect traffic volume and duration, and their label composition stays close to the overall dataset ratio, so these seven features do not cleanly separate attack traffic from benign traffic. Ground-truth profiles show that PortScan flows are nonetheless distinctive near-instant probes with very few packets. The work is delivered as a reproducible notebook pipeline and an interactive Streamlit dashboard.

2.Introduction

2.1 Background

Modern networks generate millions of flow records per day, and only a small share of them are malicious. Labelled attack data is expensive to obtain, so unsupervised learning is attractive for grouping traffic by behaviour and highlighting unusual patterns. CICIDS2017 is a widely used benchmark that provides realistic benign traffic mixed with attacks such as PortScan and DDoS, described by roughly 78 numeric features per flow produced by CICFlowMeter.

2.2 Motivation for the Project

Security analysts cannot inspect every flow. If traffic could be grouped automatically into meaningful behavioural families, triage would be faster. This project tests whether standard clustering algorithms can deliver that on real, imbalanced data, and reports clearly where they succeed and where they do not.

2.3 Overview of the Work Carried Out

The work covered loading and combining the CICIDS2017 flow files, cleaning them, selecting seven interpretable features, drawing a stratified sample, scaling the features, choosing k with elbow and silhouette analysis, running K-Means and Hierarchical clustering, evaluating both with

three internal metrics, projecting the result into 3D with PCA, and comparing clusters against the ground-truth labels. A Streamlit dashboard (TRAFFIC-INTEL) was built to explore the clusters interactively.

2.4 Scope and Responsible-Use Statement

The study uses a public, anonymised research dataset and performs offline analysis only; no live traffic was captured and no attack tooling was built. The clusters are an exploratory analysis aid, not an attack classifier, and must not be used as the sole basis for blocking users or traffic. Ground-truth labels were used only to interpret results after clustering, never to train or tune the clustering itself.

3. Problem Statement

3.1 Case-Study Scenario

A security operations team receives a large, mostly benign stream of network flow records. Labels are not available in real time. The team wants to know whether unsupervised grouping of flow statistics can reveal malicious traffic patterns such as port scans and flooding attacks.

3.2 Central Case-Study Question

Can K-Means and Hierarchical clustering, applied to a small set of flow-level features, group CICIDS2017 traffic in a way that separates malicious patterns from benign traffic?

3.3 Formal Problem Definition

Given a set of n flow records, each described by a feature vector x in R^7 after preprocessing, partition the records into k clusters without using the label column so that flows in the same cluster are similar and flows in different clusters are dissimilar. Quality is judged with internal indices (Silhouette, Davies-Bouldin, Calinski-Harabasz), and the partition is then compared with the ground-truth labels to judge its usefulness for attack detection.

3.4 Objectives that Guide the Study

1. Build a clean, reproducible preprocessing pipeline for CICIDS2017 flow data.

2. Select a compact, interpretable feature set and justify it.

3. Cluster the data with K-Means and Hierarchical clustering and choose k with evidence.

4. Compare the two algorithms using three internal validity metrics.

5. Test honestly whether the clusters align with attack and benign labels.

6. Present the results in an interactive dashboard.

3.5 Assumptions and Limitations

Flow records are treated as independent observations, because the exported files contain no timestamp column and temporal analysis was therefore not possible. Only seven features were used, so behaviour captured by other CICFlowMeter features is ignored. Clustering was run on a stratified sample rather than every flow, to keep Hierarchical clustering tractable. Internal metrics describe geometric tidiness, not whether a cluster is malicious. The dominant BENIGN class (about 83% of the data, with roughly 17% attack traffic) limits what unsupervised methods can reveal about rare attacks.

4. Dataset and Exploratory Data Analysis

4.1 About the Dataset

The dataset is CICIDS2017 from the Canadian Institute for Cybersecurity, distributed as several CSV files of bidirectional flows with about 78 numeric features and a Label column (BENIGN or an attack name). After combining the files, column names were stripped of stray whitespace. The raw combined size was 2,830,743 rows, which fell to 2,520,798 after cleaning. A stratified sample of 20,000 flows was used for clustering.

The seven features chosen for clustering are listed below.

Feature	**Meaning**
Flow Duration	Length of the flow in microseconds
Total Fwd Packets	Packets sent from source to destination
Flow Bytes/s	Throughput of the flow in bytes per second
Flow IAT Mean	Mean inter-arrival time between packets
Fwd Packet Length Mean	Average payload size of forward packets
SYN Flag Count	Number of packets with the SYN flag set
Down/Up Ratio	Ratio of download to upload traffic

4.2 Purpose of EDA

EDA was used to understand data quality, class imbalance and feature behaviour before clustering, and to decide which cleaning and scaling steps were necessary, since distance-based algorithms are sensitive to scale, skew and duplicates.

4.3 Preparation Steps Actually Performed

1. Combine and tidy. All CSV files were concatenated and column names stripped of whitespace.

2. Remove invalid values. Infinite values (common in rate features such as Flow Bytes/s) were converted to missing values and those rows dropped.

3. Deduplicate on the full feature set. Duplicate flows were removed using all 78 features before selecting the seven. An earlier version compared only the seven selected columns, which wrongly treated different flows as duplicates and almost erased PortScan from the sample. This was identified during analysis and corrected, and all reported numbers come from the corrected pipeline.

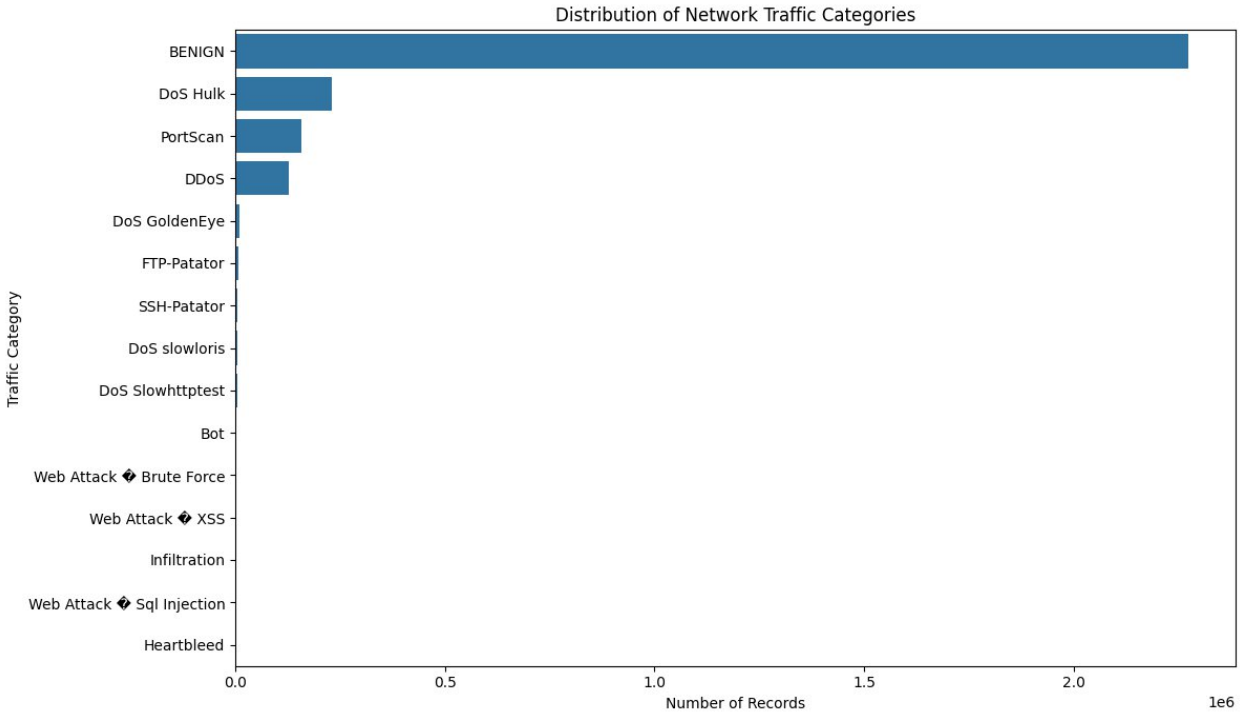
4. Feature selection. Seven interpretable flow-level features were kept (Table 1).

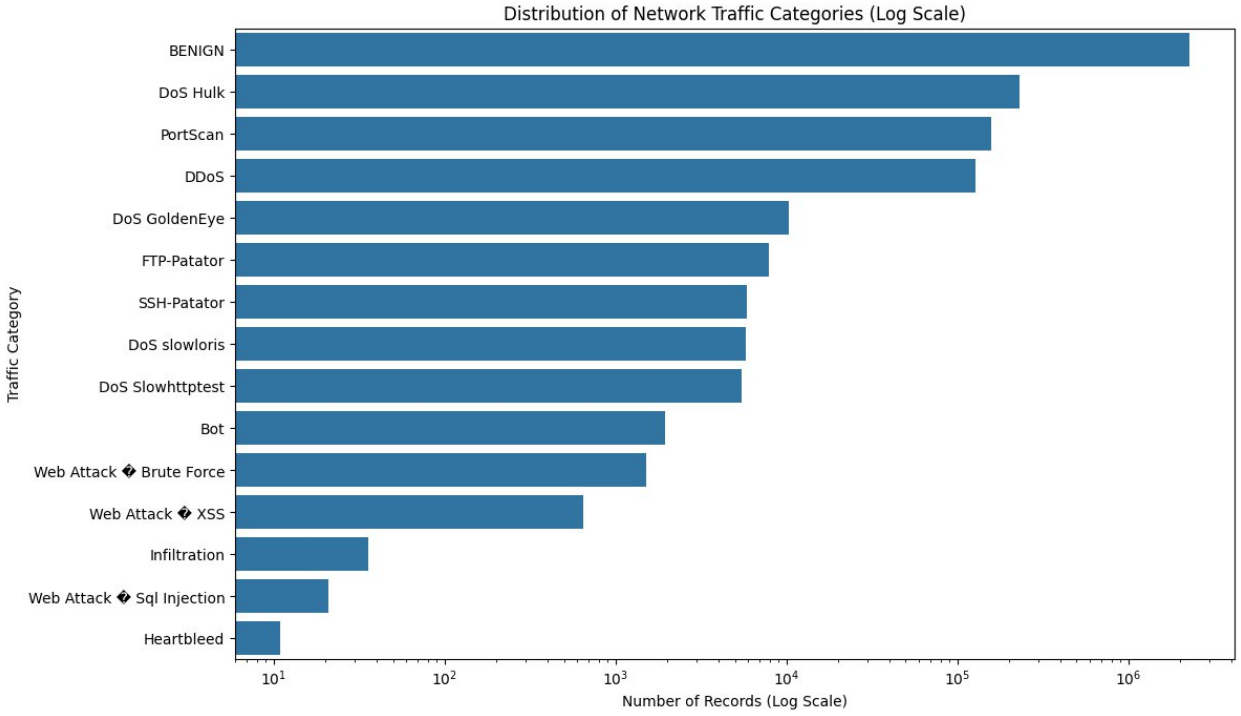
5. Stratified sampling. A sample was drawn that preserves the label proportions, so rare attacks stay represented.

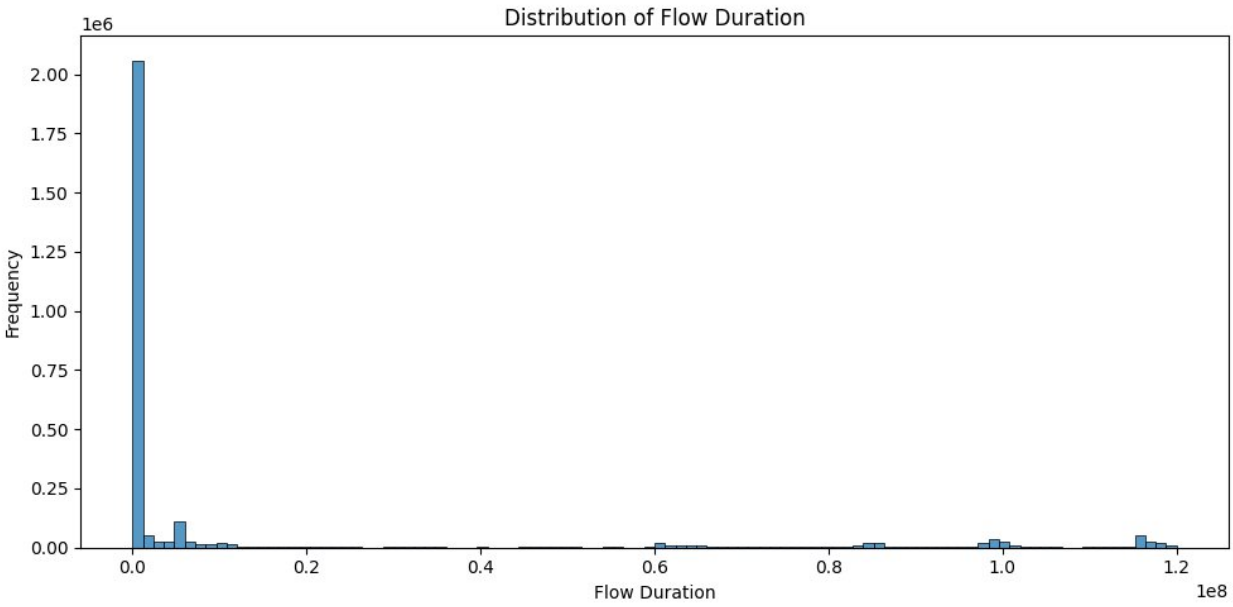
6. Scaling. Features were standardised (zero mean, unit variance) before clustering. The heavily skewed Flow Duration and Flow Bytes/s features were log-transformed first: confirm against your notebook.

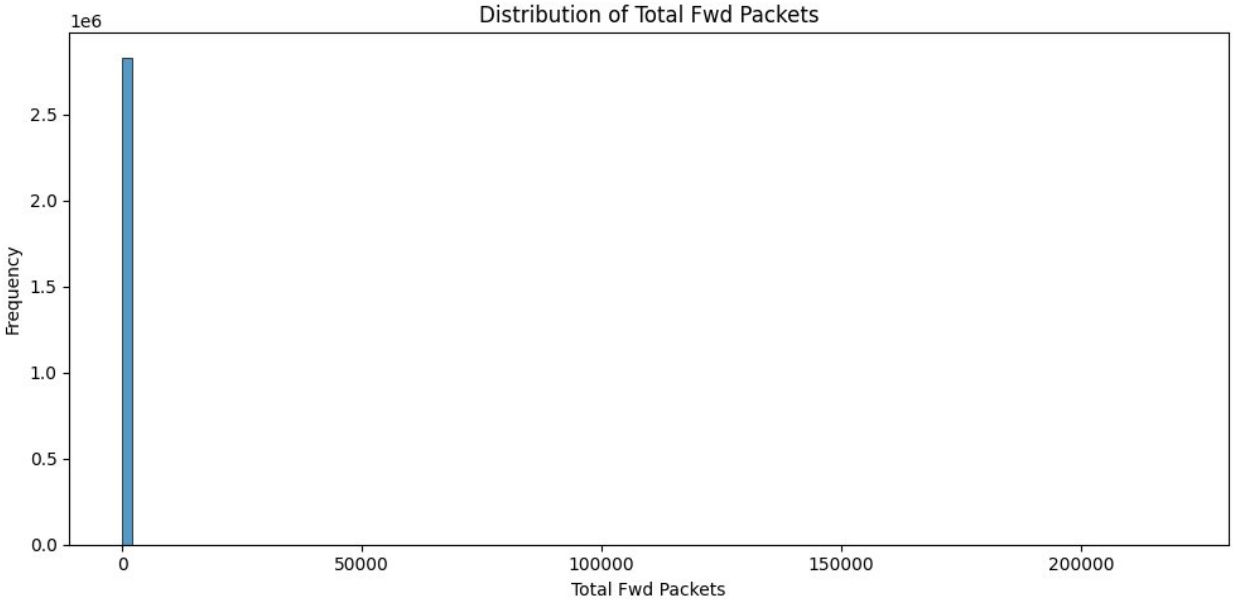
4.4 EDA Exploration - Implementation

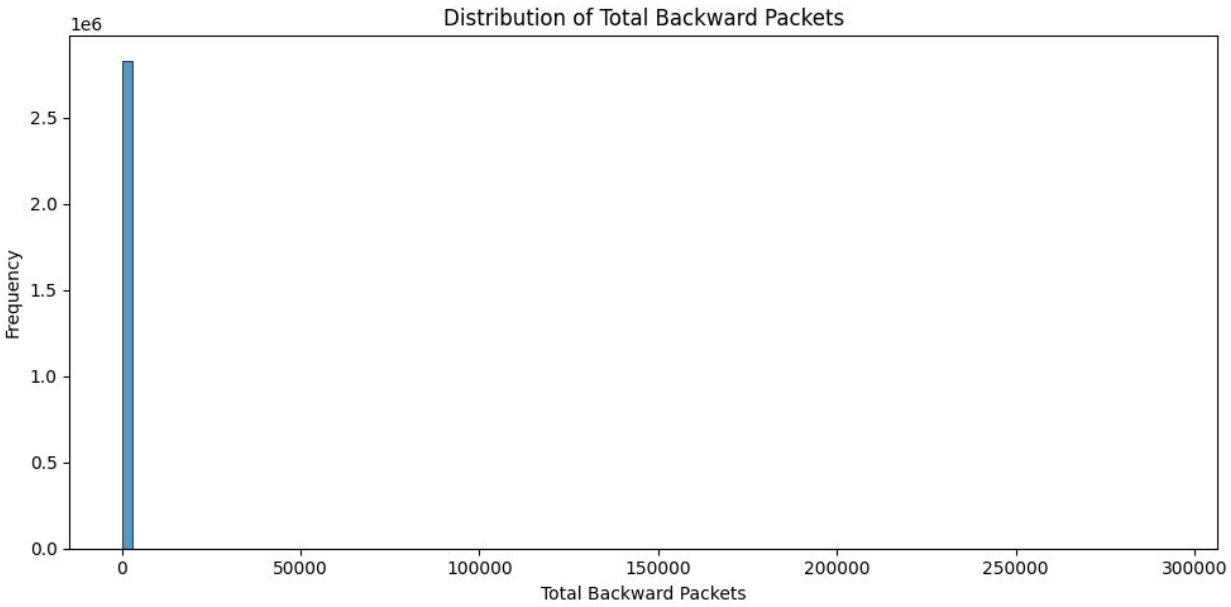
The exploration examined the distribution of traffic labels, numerical feature distributions, descriptive statistics, potential outliers and correlations among features. The analysis revealed an imbalance in traffic classes, considerable variation in feature scales, and highly skewed distributions in several network-flow features. Outlier analysis highlighted extreme values in multiple features, while correlation heatmaps and hexbin plots helped identify relationships between network-flow characteristics. These observations guided the data preprocessing, feature selection and standardisation steps before clustering.

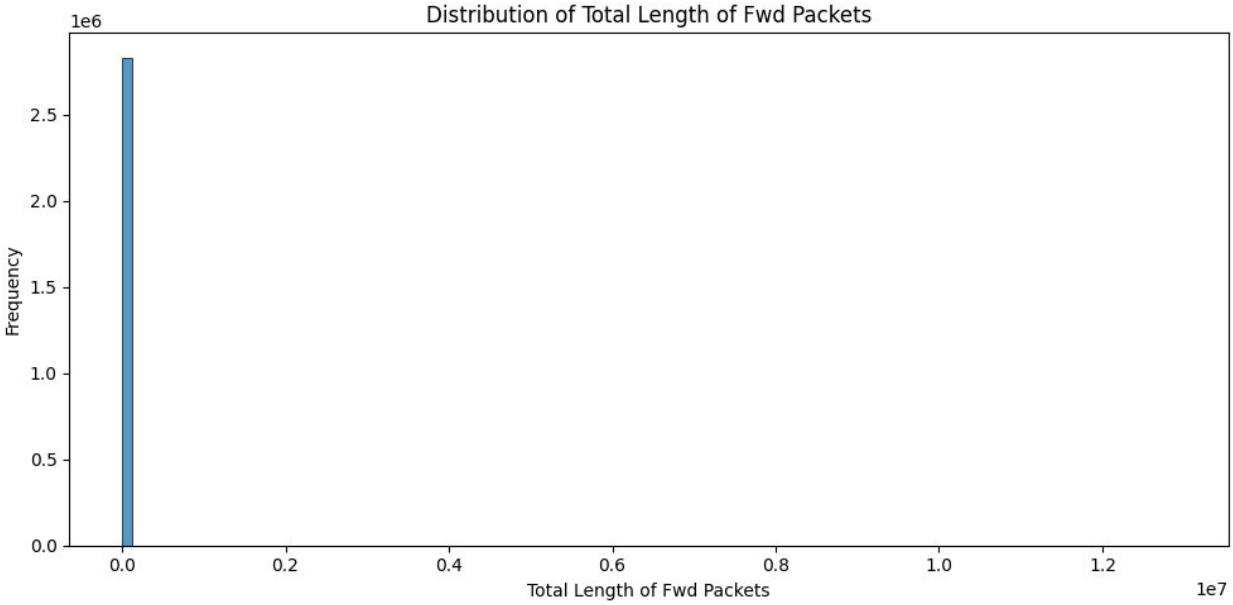

Distribution of Network Traffic Categories
BENIGN
DoS Hulk
PortScan
DDoS
DoS GoldenEye
FTP-Patator
SSH-Patator
DoS slowloris
DoS Slowhttptest
Bot
Web Attack • Brute Force
Web Attack • XSS
Infiltration
Web Attack • Sql Injection
Heartbleed
Traffic Category
0.0
0.5
1.0
1.5
2.0
Number of Records
1e6

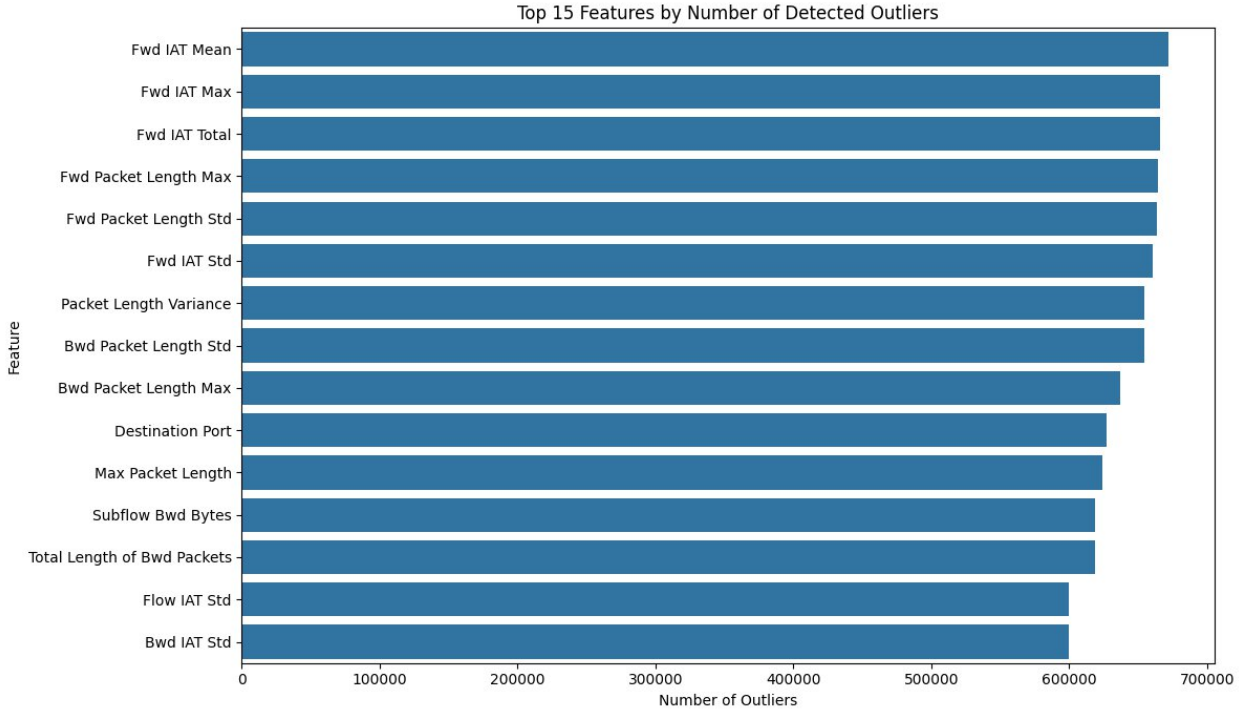

Distribution of Network Traffic Categories (Log Scale)
BENIGN
DoS Hulk
PortScan
DDoS
DoS GoldenEye
FTP-Patator
SSH-Patator
DoS slowloris
DoS Slowhttptest
Bot
Web Attack • Brute Force
Web Attack • XSS
Infiltration
Web Attack • Sql Injection
Heartbleed
Traffic Category
10^1
10^2
10^3
10^4
10^5
10^6
Number of Records (Log Scale)

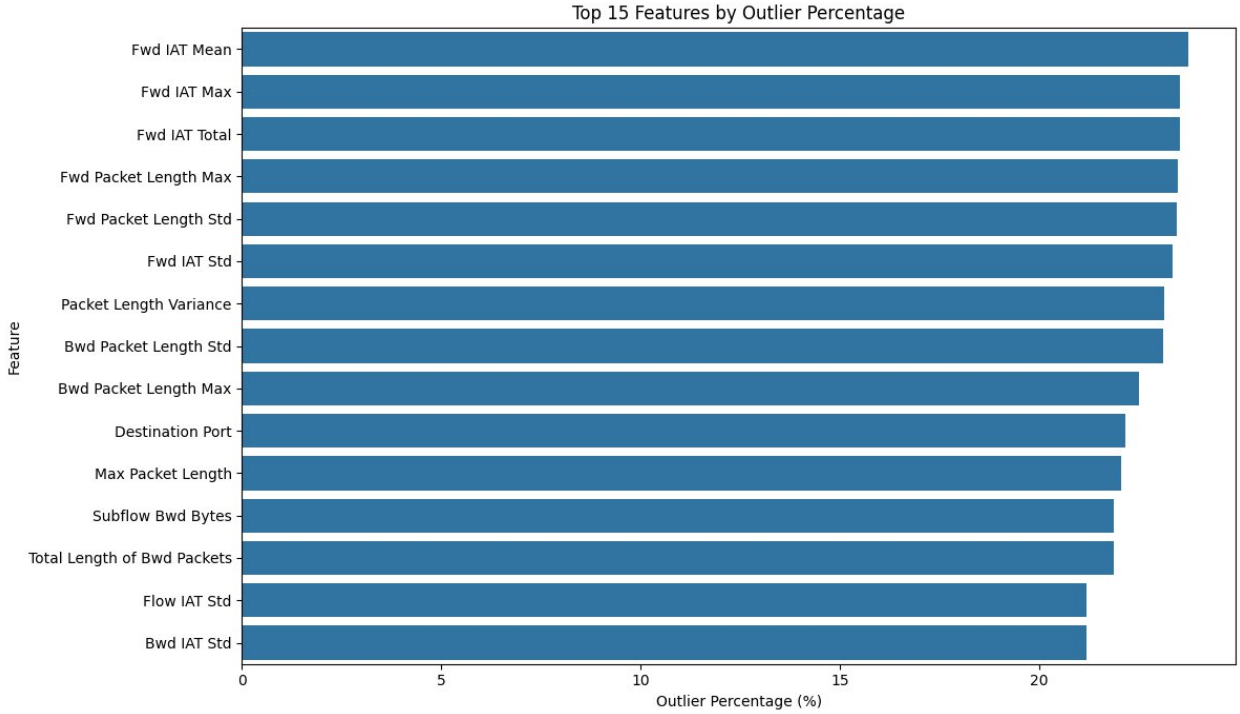

Distribution of Flow Duration
1e6
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Frequency
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Flow Duration
1e8

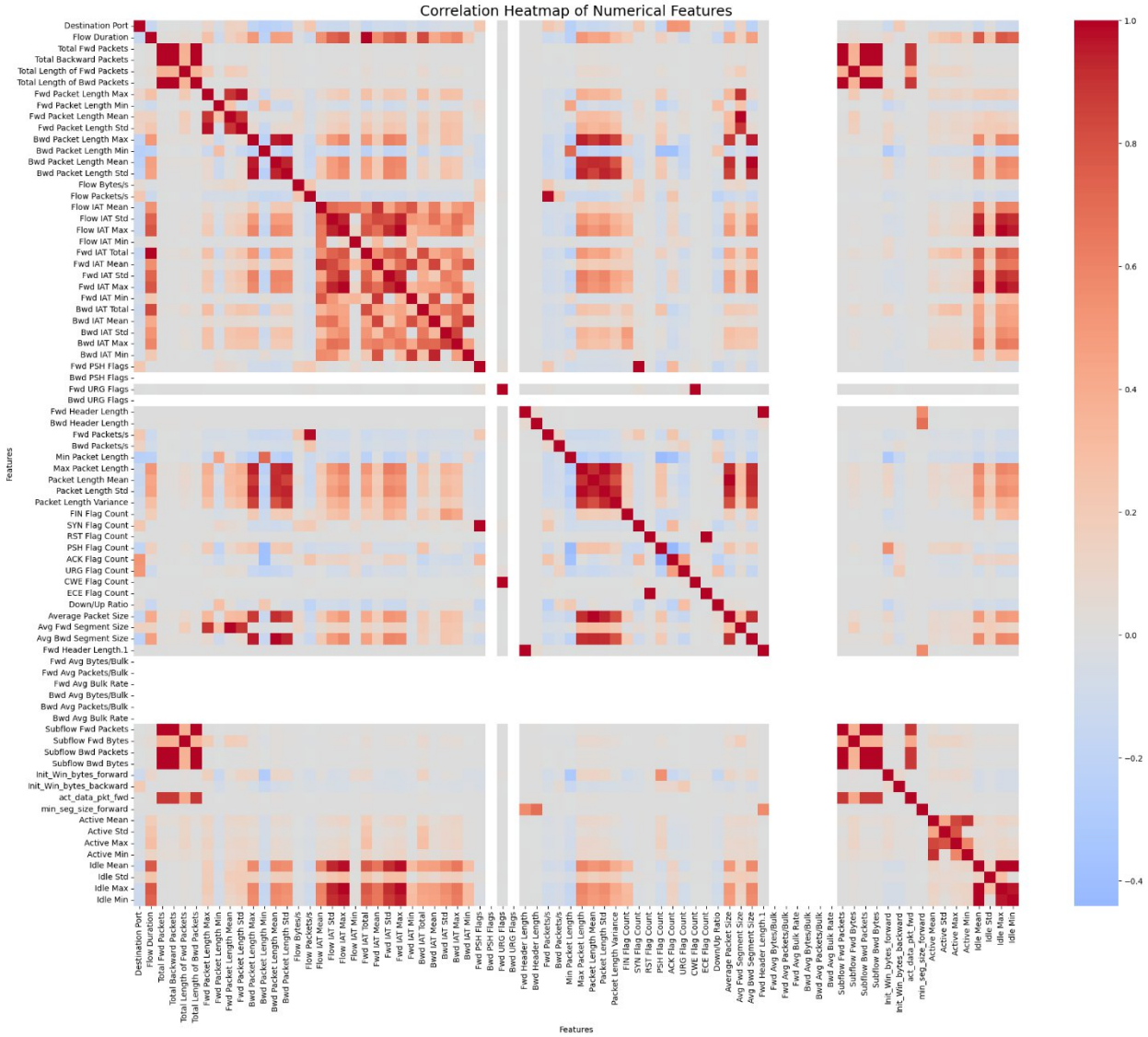

Distribution of Total Fwd Packets
1e6
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
0
50000
100000
150000
200000
Total Fwd Packets

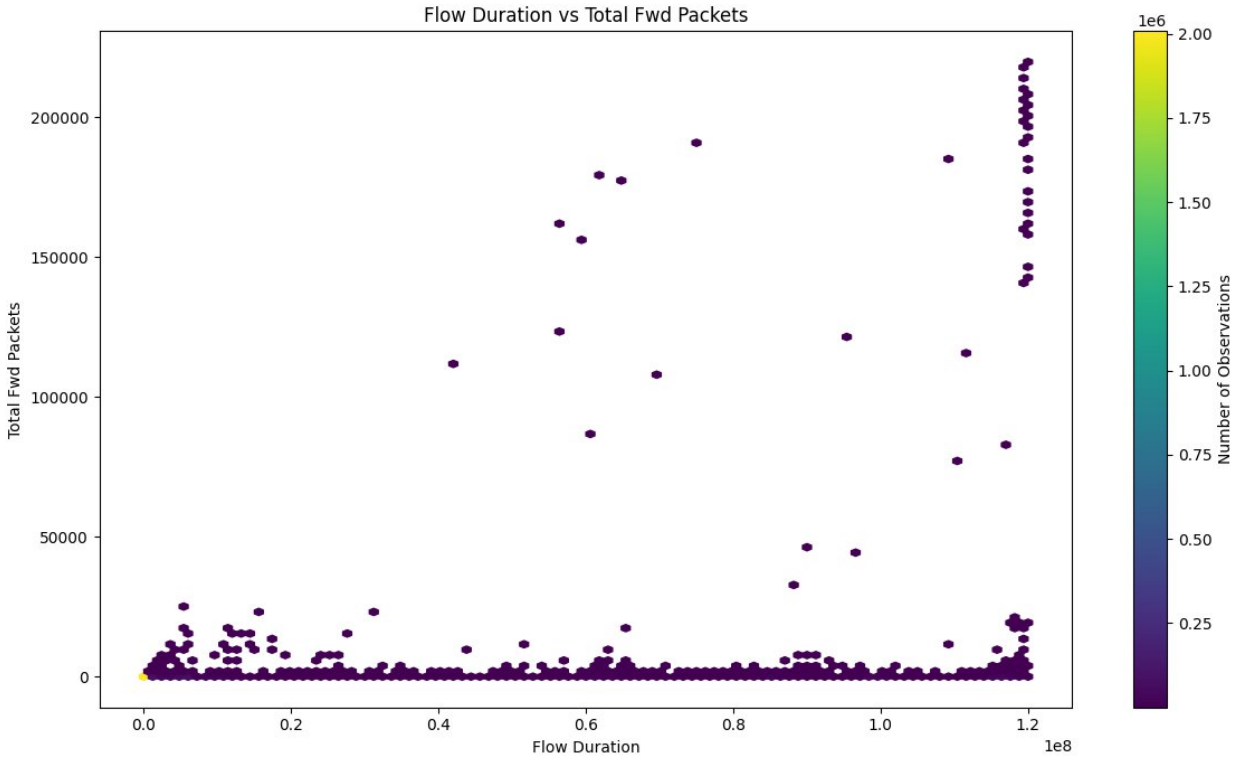

Distribution of Total Backward Packets
1e6
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
0
50000
100000
150000
200000
250000
300000
Total Backward Packets

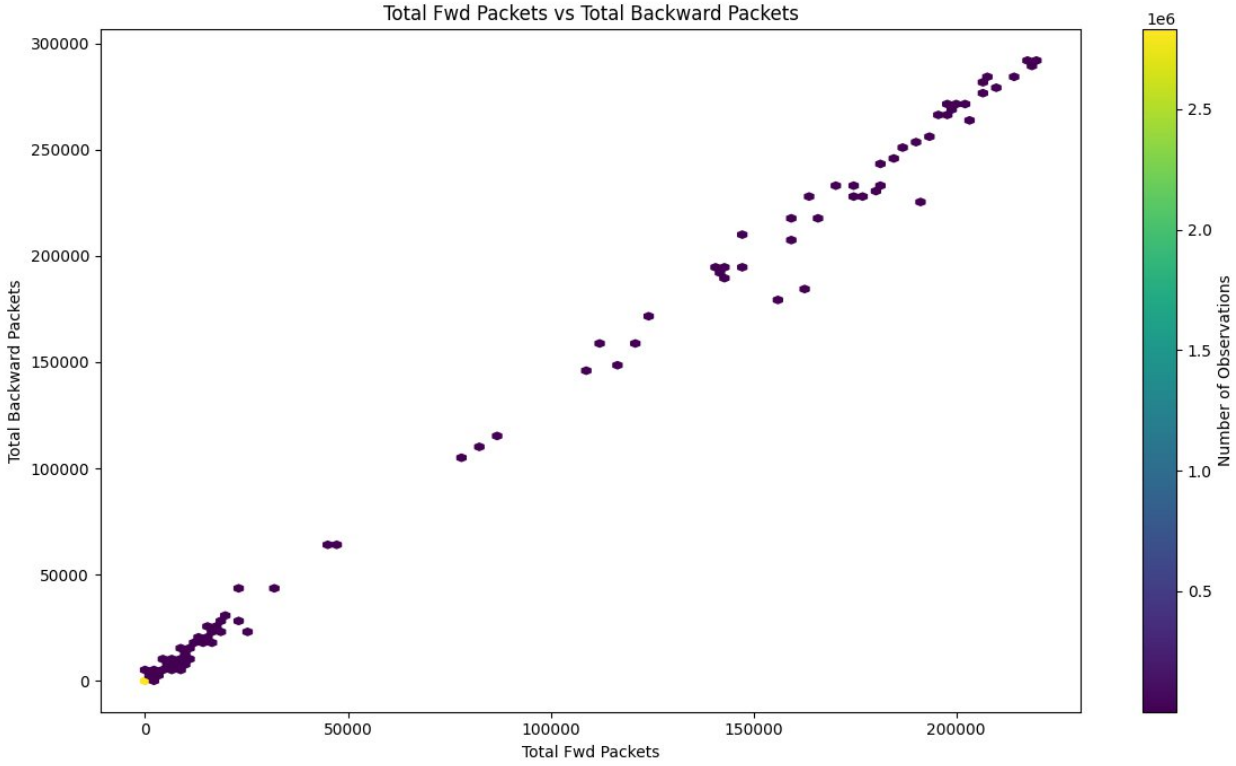

Distribution of Total Length of Fwd Packets
1e6
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1e7
Total Length of Fwd Packets

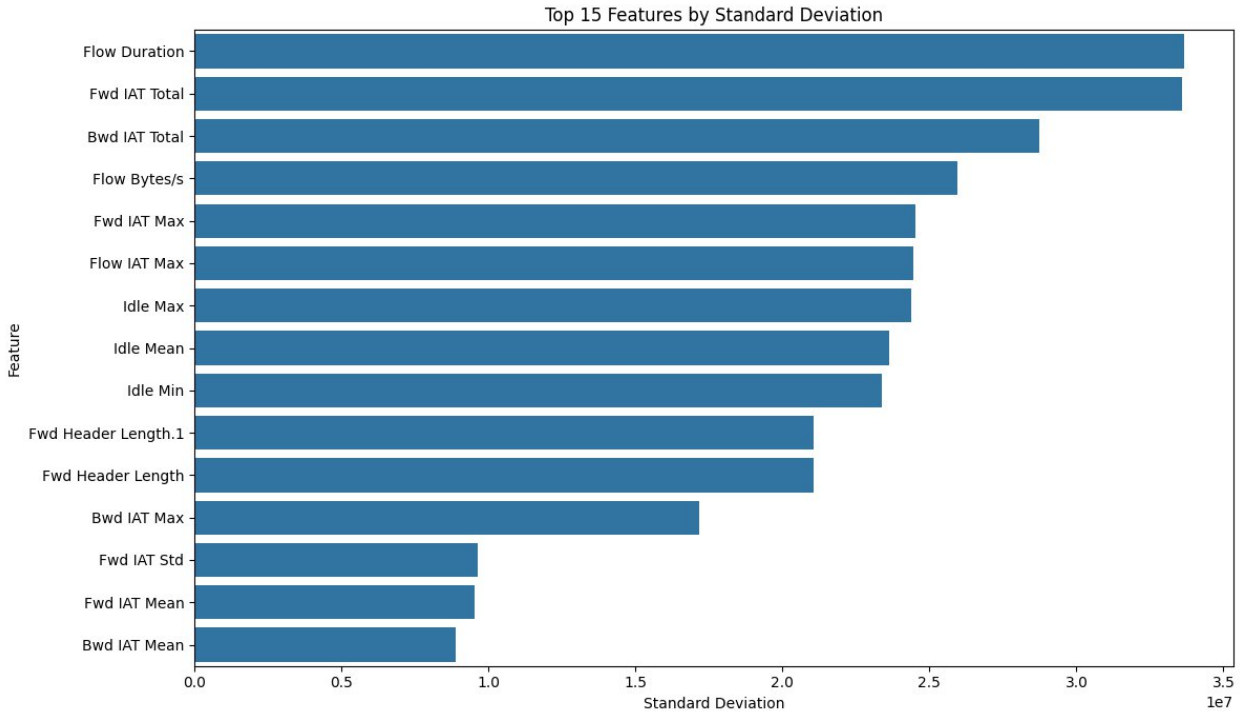

Top 15 Features by Number of Detected Outliers
Fwd IAT Mean
Fwd IAT Max
Fwd IAT Total
Fwd Packet Length Max
Fwd Packet Length Std
Fwd IAT Std
Packet Length Variance
Bwd Packet Length Std
Bwd Packet Length Max
Destination Port
Max Packet Length
Subflow Bwd Bytes
Total Length of Bwd Packets
Flow IAT Std
Bwd IAT Std
Feature
0
100000
200000
300000
400000
500000
600000
700000
Number of Outliers


Top 15 Features by Outlier Percentage
Fwd IAT Mean
Fwd IAT Max
Fwd IAT Total
Fwd Packet Length Max
Fwd Packet Length Std
Fwd IAT Std
Packet Length Variance
Bwd Packet Length Std
Bwd Packet Length Max
Destination Port
Max Packet Length
Subflow Bwd Bytes
Total Length of Bwd Packets
Flow IAT Std
Bwd IAT Std
Feature
0
5
10
15
20
Outlier Percentage (%)


Correlation Heatmap of Numerical Features
Features
Destination Port
Flow Duration
Total Fwd Packets
Total Backward Packets
Total Length of Fwd Packets
Total Length of Bwd Packets
Fwd Packet Length Max
Fwd Packet Length Min
Fwd Packet Length Mean
Fwd Packet Length Std
Bwd Packet Length Max
Bwd Packet Length Min
Bwd Packet Length Mean
Bwd Packet Length Std
Flow Bytes/s
Flow Packets/s
Flow IAT Mean
Flow IAT Std
Flow IAT Max
Flow IAT Min
Fwd IAT Total
Fwd IAT Mean
Fwd IAT Std
Fwd IAT Max
Fwd IAT Min
Bwd IAT Total
Bwd IAT Mean
Bwd IAT Std
Bwd IAT Max
Bwd IAT Min
Fwd PSH Flags
Bwd PSH Flags
Fwd URG Flags
Bwd URG Flags
Fwd Header Length
Bwd Header Length
Fwd Packets/s
Bwd Packets/s
Min Packet Length
Max Packet Length
Packet Length Mean
Packet Length Std
Packet Length Variance
FIN Flag Count
SYN Flag Count
RST Flag Count
PSH Flag Count
ACK Flag Count
URG Flag Count
CWE Flag Count
ECE Flag Count
Down/Up Ratio
Average Packet Size
Avg Fwd Segment Size
Avg Bwd Segment Size
Fwd Header Length.1
Fwd Avg Bytes/Bulk
Fwd Avg Packets/Bulk
Fwd Avg Bulk Rate
Bwd Avg Bytes/Bulk
Bwd Avg Packets/Bulk
Bwd Avg Bulk Rate
Subflow Fwd Packets
Subflow Fwd Bytes
Subflow Bwd Packets
Subflow Bwd Bytes
Init_Win_bytes_forward
Init_Win_bytes_backward
act_data_pkt_fwd
min_seg_size_forward
Active Mean
Active Std
Active Max
Active Min
Idle Mean
Idle Std
Idle Max
Idle Min
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


Flow Duration vs Total Fwd Packets
Total Fwd Packets
Flow Duration
1e8
Number of Observations
1e6


Total Fwd Packets vs Total Backward Packets
Total Backward Packets
Total Fwd Packets
Number of Observations
1e6



4.5 Summary of EDA Findings

Exploratory Data Analysis (EDA) was performed on the CICIDS2017 dataset to understand its structure, class distribution and feature characteristics. The dataset contained 2,830,743 network-flow records with 78 numerical features and one label column. The analysis revealed a significant class imbalance, with benign traffic accounting for approximately 80.30% of the records. Feature distributions showed considerable variation in numerical ranges and positive skewness. Invalid infinite values were also identified in rate-based features. Outlier analysis using the IQR method highlighted extreme values in several features, while correlation analysis and hexbin plots helped examine relationships between network-flow characteristics. Overall, the EDA highlighted the need to handle invalid values, select relevant features and standardise the data before clustering. These findings guided the preprocessing steps and helped establish a suitable foundation for applying K-Means and Hierarchical clustering.

5.Algorithm Selection and Implementation

Algorithms Chosen

K-Means partitions data into k spherical clusters by minimising within-cluster variance; it is fast and a standard baseline. **Hierarchical (Agglomerative) clustering** builds a tree of merges and was cut at k clusters, which gives a structurally different view without assuming spherical clusters as strongly. **PCA** reduced the standardised features to three components for visualisation only; the clustering itself ran in the original standardised feature space.

Choosing the Number of Clusters

Elbow and silhouette analysis were run for a range of k. The silhouette score was highest at k = 7, but inspection showed that k = 7 mostly split the dominant BENIGN class into further sub-groups without isolating attack types any better. k = 4 was therefore chosen for interpretability, and both k = 4 and k = 7 were tested. In both cases the label composition of each cluster stayed close to the overall dataset ratio.

Evaluation Results

Algorithm	Silhouette (higher better)	Davies-Bouldin (lower better)	Calinski-Harabasz (higher better)
K-Means (k=4)	0.384	0.827	5514.487
Hierarchical (k=4)	0.348	0.954	5102.269

K-Means outperformed Hierarchical clustering on all three indices. Both silhouette values are moderate, meaning clusters are only fairly well separated. The three PCA components together explain 60.4% of the total variance, so proximity in the 3D view is approximate rather than exact.

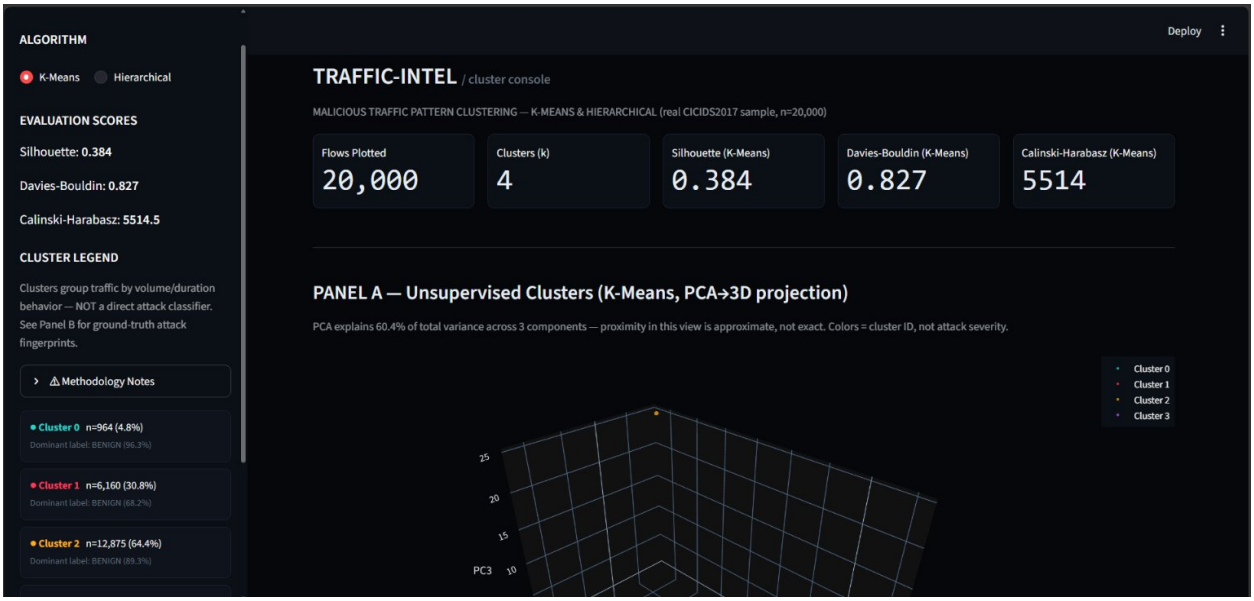
Key Finding: Clusters Reflect Volume, Not Attack Type

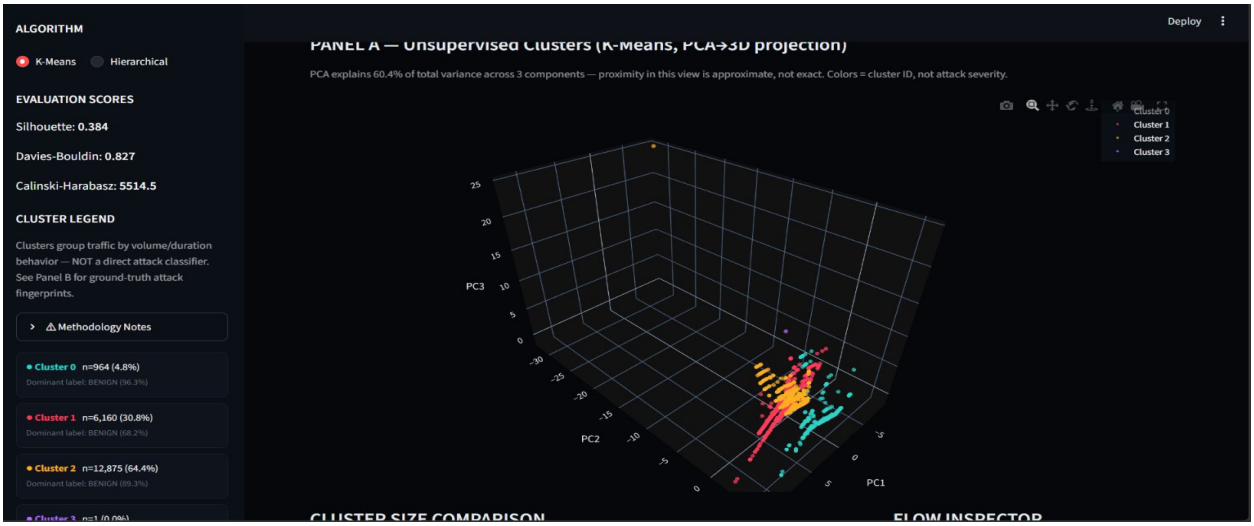
Comparing cluster membership with ground-truth labels showed that clusters group flows by volume and duration behaviour. Because every cluster contained roughly the same mix of benign and attack flows as the whole dataset, these seven features do not cleanly separate attack from benign traffic. The only attack with a clearly distinctive profile is PortScan, whose near-instant, few-packet probes stand apart in the ground-truth table, although the clustering did not isolate it on its own.

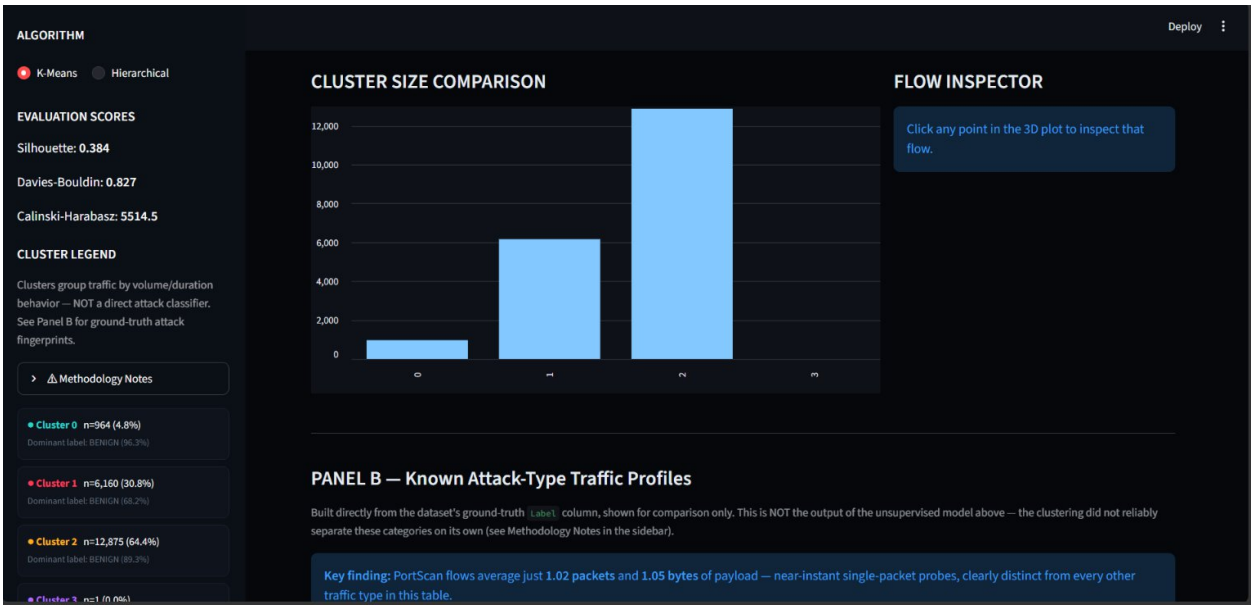
Interactive Dashboard

TRAFFIC-INTEL is a Streamlit application built on the exported clustering results. Panel A shows an interactive 3D PCA scatter plot that can switch between K-Means and Hierarchical results; clicking a point opens a Flow Inspector with the flow's feature values and its known label. Panel B shows ground-truth attack profiles, deliberately kept separate from the unsupervised output, along with the metric comparison table, a label filter and a CSV download.

TRAFFIC-INTEL Dashboard Interface









6.Conclusion

This case study built a clean, corrected pipeline for clustering CICIDS2017 flow data and compared K-Means and Hierarchical clustering at k = 4, with K-Means giving the better internal scores (Silhouette 0.384, Davies-Bouldin 0.827, Calinski-Harabasz 5514.5). The more important result is that, with these seven features, unsupervised clusters track traffic volume rather than

maliciousness and do not reliably isolate attacks, a limitation that was tested at both k = 4 and k = 7 and reported openly.

Future work includes adding DBSCAN, whose noise label may capture sparse attack traffic better than forced partitions, adding timing-regularity features such as Fwd IAT Std, using time-aware features if timestamps become available, and combining clustering with supervised classifiers for deployment.

Signature of the Team members